

EXPERIMENT 13 TO 20

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Experiment 13: Survey Results Analysis

CODE:-

```
# Experiment 13: Geographic Data
```

```
data <- data.frame(
```

```
  City = c("City A", "City B", "City C"), Population
```

```
  = c(500000, 700000, 600000),
```

```
  Temperature = c(75, 68, 80),
```

```
  Elevation = c(1000, 800, 1200),
```

```
  Longitude = c(-118.25, -74.00, -87.65), # Example Longitude values
```

```
  Latitude = c(34.05, 40.71, 41.88) # Example Latitude values
```

```
)
```

```
plot( data$Longitude,
```

```
  data$Latitude,
```

```
  main = "Geographic Distribution of Cities", xlab =
```

```
  "Longitude",
```

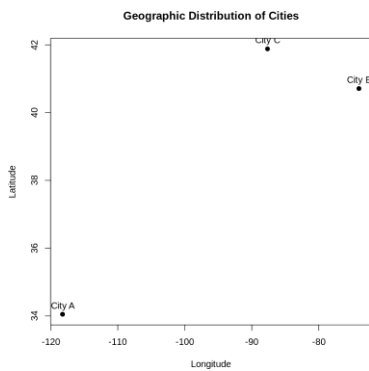
```
  ylab = "Latitude",
```

```
  pch = 19
```

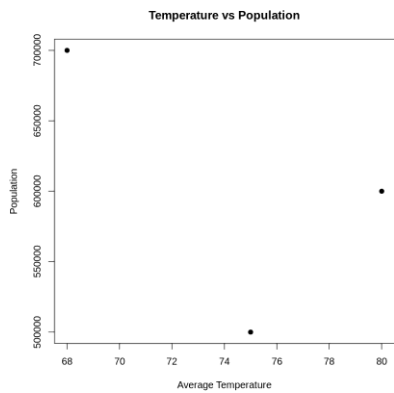
```
)
```

```
text( data$Longitude,  
      data$Latitude,  
      labels = data$City,  
      pos = 3  
)  
  
plot( data$Temperature,  
      data$Population,  
      main = "Temperature vs Population",  
      xlab = "Average Temperature",  
      ylab = "Population",  
      pch = 19  
)  
  
print(data)
```

OUTPUT :-



	City	Population	Temperature	Elevation	Longitude	Latitude
1	City A	5e+05	75	1000	-118.25	34.05
2	City B	7e+05	68	800	-74.00	40.71
3	City C	6e+05	80	1200	-87.65	41.88



Experiment 14: Survey Results Analysis

```

survey <- data.frame(
  Respondent = c(1, 2, 3), Q1
  = c("A", "B", "C"),
  Q2 = c("B", "A", "A"),
  Q3 = c("C", "D", "B")
)

barplot(table(survey$Q1),
  main = "Question 1 Responses", xlab
  = "Answers",
  ylab = "Frequency") library(fmsb)

radar <- data.frame( Q1
  = c(3, 0, 1, 1, 1, 0),
  Q2 = c(3, 0, 2, 1, 0, 0),

```

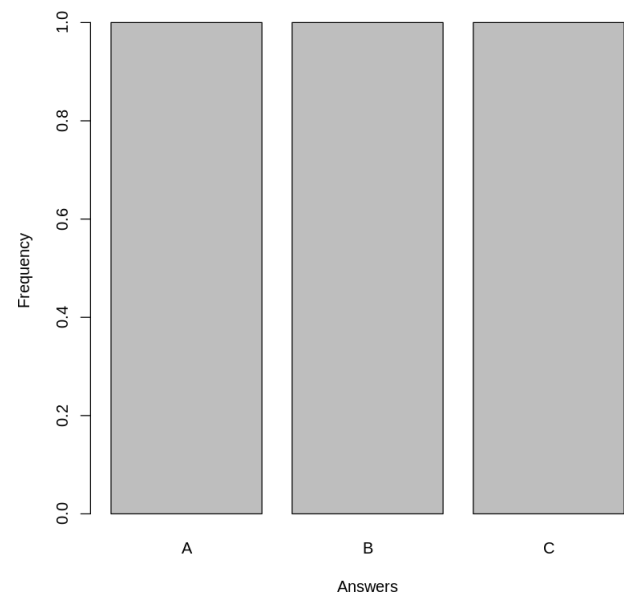
```

Q3 = c(3, 0, 1, 1, 1, 1)
)

rownames(radar) <- c("Max", "Min", "A", "B", "C", "D") radarchart(radar,
                        title = "Survey Response Pattern") survey

```

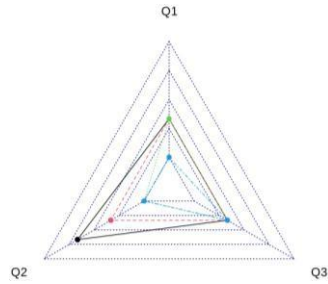
OUTPUT:-



```

# A data.frame: 3 x 4
  Respondent    Q1    Q2    Q3
    <dbl> <chr> <chr> <chr>
1         1     A     B     C
2         2     B     A     D
3         3     C     A     B

```



Experiment 15: Product Sales Analysis

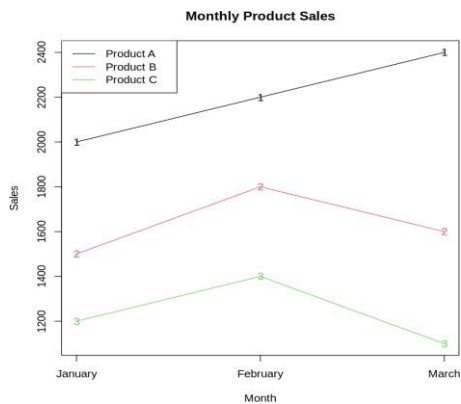
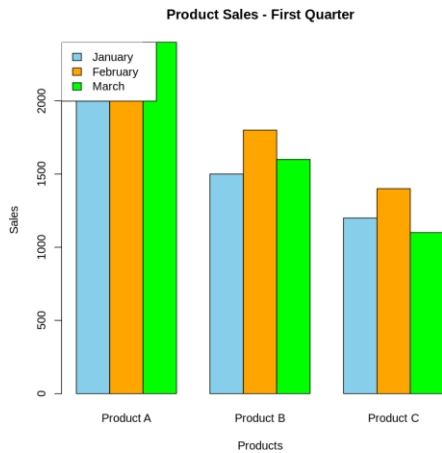
CODE:-

```
sales <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

barplot(
  t(sales[, 2:4]),
  beside = TRUE,
  names.arg = sales$Product,
  col = c("skyblue", "orange", "green"),
  main = "Product Sales - First Quarter",
  xlab = "Products",
  ylab = "Sales"
)
```

```
legend(  
  "topleft",  
  legend = c("January", "February", "March"),  
  fill = c("skyblue", "orange", "green")  
)  
months <- c("January", "February", "March")  
matplot(  
  months,  
  t(sales[, 2:4]),  
  type = "o",  
  lty = 1,  
  col = 1:3,  
  xlab = "Month",  
  ylab = "Sales",  
  main = "Monthly Product Sales"  
)  
legend(  
  "topleft",  
  legend = sales$Product,  
  col = 1:3,  
  lty = 1  
)  
print(sales)
```

OUTPUT :-



	Product	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

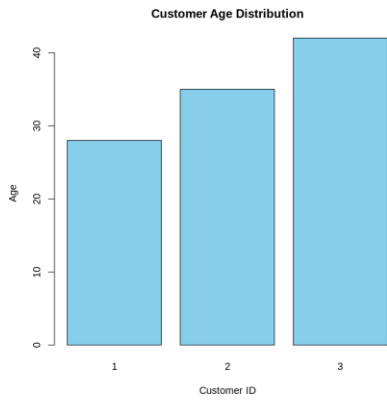
Experiment 16: Customer Demographics Analysis

CODE :-

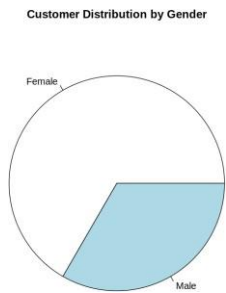
```
customer <- data.frame(  
  ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),
```

```
Income = c(50000, 60000, 75000)
)
barplot(
  customer$Age,
  names.arg = customer$ID,
  col = "skyblue",
  main = "Customer Age Distribution",
  xlab = "Customer ID",
  ylab = "Age"
)
gender <- table(customer$Gender)
pie(
  gender,
  labels = names(gender),
  main = "Customer Distribution by Gender"
)
print(customer)
```

OUTPUT:-



ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000



Experiment 17: Employee Performance Analysis

CODE:-

```
employee <- data.frame(
  ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years = c(5, 3, 7),
  Performance = c(85, 92, 78)
)
```

```

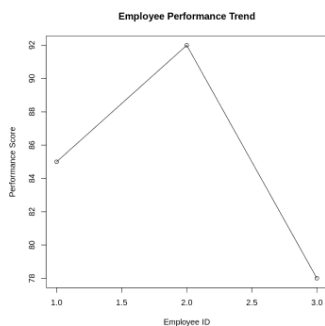
plot(
  employee$ID,
  employee$Performance,
  type = "o",
  main = "Employee Performance Trend",
  xlab = "Employee ID",
  ylab = "Performance Score"
)

barplot(
  table(employee$Department),
  col = "skyblue",
  main = "Employees by Department",
  xlab = "Department",
  ylab = "Number of Employees"
)

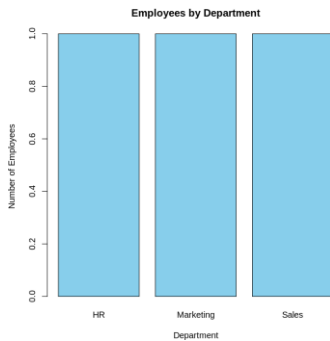
print(employee)

```

OUTPUT:-



	ID	Department	Years	Performance
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78



Experiment 18: Product Inventory Management

CODE:-

```
inventory <- data.frame(
  ID = c(1, 2, 3),
  Product = c("Product A", "Product B", "Product C"),
  Quantity = c(250, 175, 300)
)

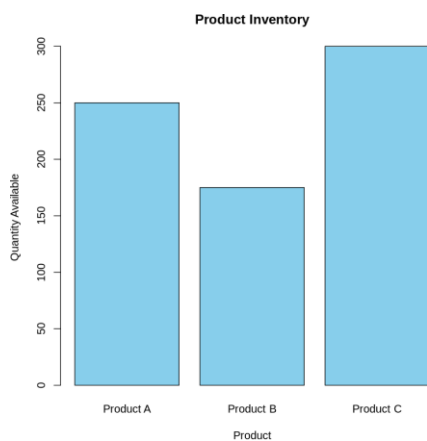
barplot(
  inventory$Quantity,
  names.arg = inventory$Product,
  col = "skyblue",
  main = "Product Inventory",
  xlab = "Product",
```

```

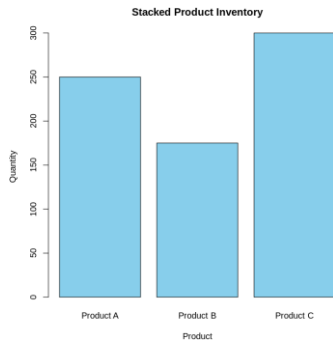
ylab = "Quantity Available"
)
barplot(
  matrix(inventory$Quantity, nrow = 1),
  names.arg = inventory$Product,
  col = c("skyblue", "orange", "green"),
  main = "Stacked Product Inventory",
  xlab = "Product",
  ylab = "Quantity"
)
print(inventory)

```

OUTPUT:-



ID	Product	Quantity
1	1 Product A	250
2	2 Product B	175
3	3 Product C	300



Experiment 19: Survey Responses Analysis

CODE:-

```
survey <- data.frame(  
  ID = c(1, 2, 3),  
  Q1 = c("A", "B", "C"),  
  Q2 = c("B", "A", "A"),  
  Q3 = c("C", "D", "B")  
)  
  
barplot(  
  table(survey$Q1),  
  col = "skyblue",  
  main = "Question 1 Responses",  
  xlab = "Answer",  
  ylab = "Frequency"  
)
```

```

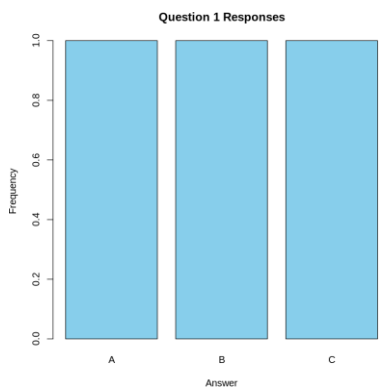
responses <- rbind(
  table(survey$Q1),
  table(survey$Q2),
  table(survey$Q3)
)

barplot(
  responses,
  beside = FALSE,
  col = c("skyblue", "orange", "green"),
  main = "Overall Survey Responses",
  xlab = "Questions",
  ylab = "Frequency"
)

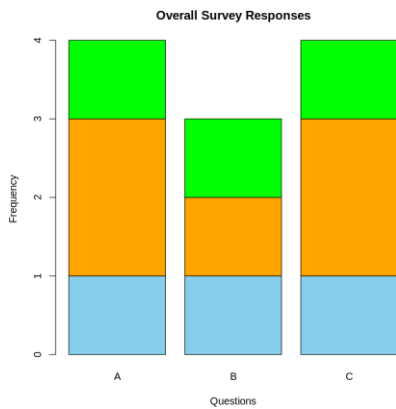
print(survey)

```

OUTPUT:-



	ID	Q1	Q2	Q3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B



Experiment 20: Stock Analysis

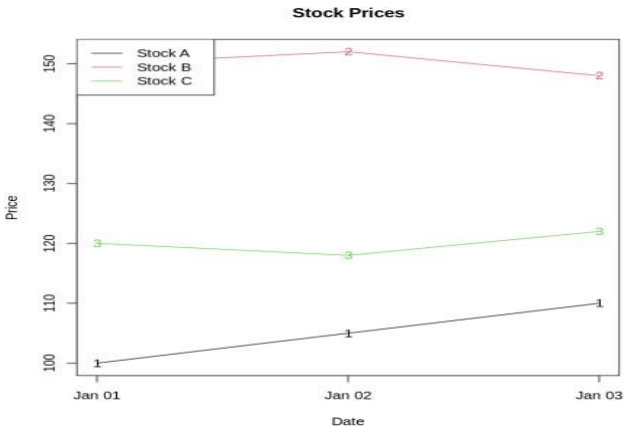
CODE:-

```
stock <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

matplot(
  stock$Date,
  stock[, 2:4],
  type = "o",
  lty = 1,
  col = 1:3,
```

```
main = "Stock Prices",
xlab = "Date",
ylab = "Price"
)
legend(
  "topleft",
  legend = c("Stock A", "Stock B", "Stock C"),
  col = 1:3,
  lty = 1
)
change <- c(
  NA,
  diff(stock$Stock_A) / stock$Stock_A[-nrow(stock)] * 100
)
barplot(
  change[-1],
  names.arg = as.character(stock$Date[-1]),
  col = "skyblue",
  main = "Daily Percentage Change - Stock A",
  xlab = "Date",
  ylab = "Percentage Change"
)
print(stock)
```


OUTPUT:-



	Date	Stock_A	Stock_B	Stock_C
1	2023-01-01	100	150	120
2	2023-01-02	105	152	118
3	2023-01-03	110	148	122

